

Methods of Violating Bell's Inequality in The Quantum and Classical Case

Isabelle Rose*

MIT Department of Mathematics

(Dated: January 5, 2026)

This paper seeks to analyse the history of Bell inequalities and the accuracy of the conclusions drawn from experiments that have successfully violated it. The primary goal is to present mechanisms by which Bell inequalities may be violated that do not appeal to or rely on lack of determinism, realism, or locality.

The strategy is to show first that quantum mechanical predictions and experimental results do not violate a corrected Bell inequality, and that there is a method of constructing a different probability measure than the one Bell attributed as necessary for local hidden variable theories, which not only reproduce the predictions of quantum mechanics, but require no abandonment of determinism or local realism.

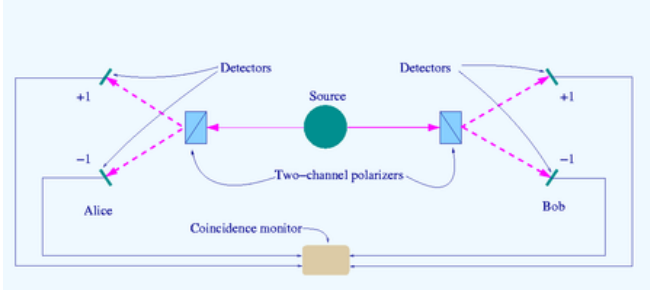


FIG. 1. Diagram of a photon analyser, a kind of apparatus used in an experiment to test if the laws of nature obey Bell inequalities. From [1]

I. DERIVATION OF BELL INEQUALITY WITH HIDDEN VARIABLES WITH MINIMAL BELL PREMISES

For the sake of recreating the obscurity that frequently surrounds this topic, I will derive a commonly used Bell inequality, and only after its derivation reveal the premises it rests on. The derivation being presented in this order hopefully sympathises the reader to the conclusive air of this sort of impossibility proof. As will be seen, the subtleties of the claim of Bell inequalities lie not in their reasoning, but in their setup.

The experimental setup is as follows. There are two measurement devices, one labelled A, another labelled B. Every measurement either device makes is either +1 or -1. These devices, we suppose, are deterministic, even if a lot of information may be required to predict their outcomes. What is known is that the outcome of a measurement device depends on its measurement setting, and other information we will classify as a hidden variable λ , which may be a scalar parameter, a function, or perhaps a mathematical object even more complicated.

Succinctly, $A(a, \lambda) \in \{+1, -1\}$ and $B(b, \lambda) \in \{+1, -1\}$. Now let us define the expected value over hid-

den variables of the measurement outcomes at A when it has measurement setting a as:

$$\langle A(a) \rangle := \int_{\Lambda} A(a, \lambda) \cdot \rho(\lambda) d\lambda \quad (1)$$

For succinctness later when I am using different measurement settings which will be denoted by $\bar{a}$ and $\bar{b}$, I will be writing $\langle A \rangle$ and $\langle \bar{A} \rangle$ which will be meant to mean $\langle A(a) \rangle$ and $\langle A(\bar{a}) \rangle$. This will hopefully be clear as there is no measurement device $\bar{A}$.

Now consider the product of the two measurement outcomes $A(a, \lambda) \cdot B(b, \lambda)$. The expected value over hidden variable of that product is:

$$\langle AB \rangle := \int_{\Lambda} A(a, \lambda) \cdot B(b, \lambda) \cdot \rho(\lambda) d\lambda \quad (2)$$

Following those definitions the derivation of the Bell inequality that will be examined in this paper is thus:

$$\begin{aligned} \langle AB \rangle - \langle A\bar{B} \rangle &= \int_{\Lambda} A(a, \lambda) \cdot B(b, \lambda) \cdot \rho(\lambda) d\lambda \quad (3) \\ &\quad - \int_{\Lambda} A(a, \lambda) \cdot B(\bar{b}, \lambda) \cdot \rho(\lambda) d\lambda \end{aligned}$$

$$\begin{aligned} \langle AB \rangle - \langle A\bar{B} \rangle &= \int_{\Lambda} A(a, \lambda) \cdot B(b, \lambda) \cdot \rho(\lambda) d\lambda \quad (4) \\ &\quad \pm \int_{\Lambda} A(a, \lambda) \cdot B(b, \lambda) \cdot A(\bar{a}, \lambda) \cdot B(\bar{b}, \lambda) \cdot \rho(\lambda) d\lambda \\ &\quad \mp \int_{\Lambda} A(a, \lambda) \cdot B(b, \lambda) \cdot A(\bar{a}, \lambda) \cdot B(\bar{b}, \lambda) \cdot \rho(\lambda) d\lambda \\ &\quad - \int_{\Lambda} A(a, \lambda) \cdot B(\bar{b}, \lambda) \cdot \rho(\lambda) d\lambda \end{aligned}$$

$$\langle AB \rangle - \langle A\bar{B} \rangle = \quad (5)$$

$$\begin{aligned} &\int_{\Lambda} A(a, \lambda) \cdot B(b, \lambda) \cdot (1 \pm A(\bar{a}, \lambda) \cdot B(\bar{b}, \lambda)) \cdot \rho(\lambda) d\lambda \\ &- \int_{\Lambda} A(a, \lambda) \cdot B(\bar{b}, \lambda) \cdot (1 \pm A(\bar{a}, \lambda) \cdot B(b, \lambda)) \cdot \rho(\lambda) d\lambda \end{aligned}$$

* ambre@mit.edu

The next step is to take the absolute value of both sides. By triangle inequality, the absolute value can be moved inside of the integral. Further notice that the quantity AB always has magnitude 1, regardless of its measurement settings or outcome, and thus disappears under absolute value. Notice additionally that the quantities $1 \pm AB$ and $\rho(\lambda)$ are never negative, and so remain unaffected by the absolute value. What we are left with is:

$$\begin{aligned} | \langle AB \rangle - \langle \bar{A}\bar{B} \rangle | &\leq \\ \int_{\Lambda} (1 \pm A(\bar{a}, \lambda) \cdot B(\bar{b}, \lambda)) \cdot \rho(\lambda) d\lambda \\ + \int_{\Lambda} (1 \pm A(\bar{A}, \lambda) \cdot B(b, \lambda)) \cdot \rho(\lambda) d\lambda \end{aligned} \quad (6)$$

$$| \langle AB \rangle - \langle \bar{A}\bar{B} \rangle | \leq 2 \pm (\langle \bar{A}\bar{B} \rangle + \langle \bar{A}B \rangle) \quad (7)$$

This is the Bell inequality we will be examining. It is actually a pair of inequalities, the left hand side having to be lesser than both right hand sides according to the derivation.

It is well known that in Quantum Mechanics, if the measurement devices A and B are measuring whether the spins of two singlet state entangled spin- $\frac{1}{2}$ particles align themselves to be parallel or anti-parallel to the measurement angles θ_a and θ_b , then the expected value of the quantity $\langle AB \rangle$ as predicted by theory and matched by experiment, is $-\cos(\theta_a - \theta_b)$

However, using the selected measuring angles of

$$\theta_a = 0 \quad \theta_b = \frac{\pi}{8} \quad \theta_{\bar{a}} = \frac{\pi}{4} \quad \theta_{\bar{b}} = \frac{3\pi}{8}$$

One arrives at the result that the left hand inequality is calculated to be 0.5412... when the right hand sides are calculated to be 0.1522... and 3.8478.... The Bell inequality is easily and largely violated. This result was experimentally confirmed for a wide range of values in Aspect 1991 [2]. The question one initially poses oneself is where did the mathematics err? Surely nature cannot violate mathematics. But it is quickly confirmed that no step in the derivation is incorrect or invalid. The cause of the Bell violation is not in the failure of mathematics, but in the over-constraining of nature that goes into writing down not even the first line, but the definition of expected value in equation 2.

The Bell premises that go into this definition of expected value I assess are as follows:

1. Determinability: The measurement devices are determined by some natural information and process. They are deterministic and any stochastic modelling is a shortcoming on our part and not an aspect of nature.
2. Instant Measurement: The outcome of the measurement devices depends only on the state of their

measurement settings and hidden variables at the instant of measurement.

3. Locality: The measurement outcome of device A does not depend on either the measurement outcome of measuring setting of device B, and vice versa.
4. Measurement Independence: The hidden variable(s) that determines the measurement outcome of the devices do not depend on the measurement settings of either device.
5. Measurement Integrity: When a measurement is made at a device, it does not affect the outcomes of future measurements made at the same device in a way to break the determinability of the hidden variables.
6. Realism: Every quantity used in the measurement process has a defined value that exists even if measurement has not taken place, most particularly the measurement settings, hidden variables, and measurement outcomes.

II. THE QUANTUM VIOLATION AND WHAT IT DOESN'T TELL US

There is a lot to discuss regarding these premises. First and foremost, it should be no surprise to anyone who views the Quantum Mechanics as an accurate theory with true axioms that it manages to violate Bell Inequalities. Quantum Mechanics disavows the axioms of: Determinability, Locality, Measurement Integrity, and Realism. And Standard Model rejects all these except Locality.

With these premises, the rejection of Measurement Integrity causes a problem. Unlike rejecting Determinability and Realism, which leave the above derivation mathematically valid but with an incorrect beginning, rejecting Measurement Integrity breaks the mathematical validity of the derivation.

If solely Determinability and Realism were rejected, and given how tied together the two premises are, one could make the argument that if they are the only premises on which one is willing to philosophically compromise, and their being true would mathematically require nature to obey Bell Inequalities, it must be the case that they are false. In fact, this is the claim made by physicists, it is the reason so much emphasis is put on Bell inequalities and their violations. [3]

Rejecting Measurement Integrity, however, means measurements made at the same device cannot commute, which is what occurs in factorisation in line 5. Rejecting Measurement Integrity, therefore, makes the above derivation invalid, and so as it stands, nature's violation of this Bell Inequality would be no cause to abandon Determinability and Realism. However, there is a correc-

tion one can make to the above derivation, as proposed by Pena-Cetto [4].

The principle is to add the premise that when a measurement is performed, it affects future measurements in a way that is still predictable. Essentially, the performance of a measurement affect future measurements in a way that is analogous to if the second measurement were the only performed, but the hidden variables governing it were different. Different in the way that corrects to how the measurement device would measure as a consequence of there in fact having already been a measurement. Mathematically, here are the rules that govern this correction:

$$\langle A \rangle = \int_{\Lambda} A(a, \lambda) M_a \rho(\lambda) d\lambda \quad (8)$$

$$\langle BA \rangle = \int_{\Lambda} B(b, \lambda) M_b A(a, \lambda) M_a \rho(\lambda) d\lambda \quad (9)$$

$$\int_{\Lambda} M \rho(\lambda) d\lambda = \int_{\Lambda} \rho(\lambda) d\lambda = 1 \quad (10)$$

$$\int_{\Lambda} M A(a) M_a \rho(\lambda) d\lambda = \int_{\Lambda} A(a) M_a \rho(\lambda) d\lambda \quad (11)$$

These rules state that, before a measurement outcome is made, the hidden variables' probability distribution function is updated by an operator (which very well may be the identity operator). Order matters when taking the expected value of product of measurement outcomes, however since by the locality assumptions everything with an A commutes with anything with a B , it is still the case that

$$\begin{aligned} \langle BA \rangle &= \int_{\Lambda} B(b, \lambda) M_b A(a, \lambda) M_a \rho(\lambda) d\lambda \quad (12) \\ &= \int_{\Lambda} M_b M_a B(b, \lambda) A(a, \lambda) \rho(\lambda) \\ &= \int_{\Lambda} B(b, \lambda) A(a, \lambda) \rho(\lambda) \end{aligned}$$

And finally, changing the pdf of the random variables once all the measurements have been made does not affect the value of the integrals over hidden variables.

With this correction, we begin the Bell derivation with the line:

$$\begin{aligned} \langle AB \rangle - \langle A\bar{B} \rangle &= \quad (13) \\ &\int_{\Lambda} A(a, \lambda) M_a B(b, \lambda) M_b \rho(\lambda) d\lambda \\ &- \int_{\Lambda} A(a, \lambda) M_a B(\bar{b}, \lambda) M_{\bar{b}} \rho(\lambda) d\lambda \end{aligned}$$

which factorises similarly to 5 into:

$$\begin{aligned} \langle AB \rangle - \langle A\bar{B} \rangle &= \quad (14) \\ &\int_{\Lambda} (1 \pm A(\bar{a})B(\bar{b})) M_{\bar{a}} M_{\bar{b}} A(a)B(b) M_a M_b d\lambda \\ &- \int_{\Lambda} (1 \pm A(\bar{a})B(b)) M_{\bar{a}} M_b A(a)B(\bar{b}) M_a M_{\bar{b}} d\lambda \end{aligned}$$

When the absolute value is taken, a lot simplifies still, but the main new artefact that does not simplify is the order of the M operators:

$$\begin{aligned} |\langle AB \rangle - \langle A\bar{B} \rangle| &= 2 \quad (15) \\ &\pm \left(\int_{\Lambda} A(\bar{a})B(\bar{b}) M_{\bar{a}} M_{\bar{b}} M_a M_b d\lambda \right. \\ &\left. + \int_{\Lambda} A(\bar{a})B(b) M_{\bar{a}} M_b M_a M_{\bar{b}} d\lambda \right) \end{aligned}$$

These integrals would be equal to $\langle \bar{A}\bar{B} \rangle - \langle \bar{A}B \rangle$ if the uncorresponding M operators could be moved to the left of the integral. However inability to commute prevents this. Instead, the interpretation is that in the first integrals, before measurements were made with settings $\bar{a}$ and $\bar{b}$, they were made with settings a and b , and identically for the second integral. In that case we can rewrite this line as:

$$\begin{aligned} |\langle AB \rangle - \langle A\bar{B} \rangle| &= \quad (16) \\ 2 \pm (\langle \bar{A}\bar{B}|AB \rangle + \langle \bar{A}B|A\bar{B} \rangle) \end{aligned}$$

The intriguing aspect of this result is that it still gives a perfectly measurable Bell Inequality, another that can be experimentally verified. It does bring up an enormous problem to those who would want to still use Bell violations to justify the extreme philosophical premises of Quantum Mechanics and the Standard Model. At least in the case of entangled singlet particles, Quantum Mechanics does give a formula for $\langle \bar{A}\bar{B}|AB \rangle$, which is:

$$\langle A(\bar{a})B(\bar{b})|A(a)B(b) \rangle = -\cos \theta_{ab} \cos \theta_{a\bar{a}} \cos \theta_{b\bar{b}} \quad (17)$$

where $\theta_{ab} \equiv \theta_a - \theta_b$.

Using this formula, this transforms the Bell inequality into:

$$\begin{aligned} |\cos \theta_{ab} - \cos \theta_{a\bar{b}}| &\leq \quad (18) \\ 2 \pm (\cos \theta_{ab} - \cos \theta_{a\bar{b}}) \cos \theta_{a\bar{a}} \cos \theta_{b\bar{b}} \end{aligned}$$

The issue is that this inequality, unlike the last, holds for all conceivable angles θ_a , θ_b , $\theta_{\bar{a}}$, $\theta_{\bar{b}}$. In which case experiment can obey this inequality, for which rejecting Measurement Integrity suffices entirely and there is no philosophical justification to reject Determinability or Realism, or the inequality is violated,

and thus the prediction of Quantum Mechanics that $\langle A(\bar{a})B(\bar{b})|A(a)B(b) \rangle = -\cos\theta_{ab}\cos\theta_{a\bar{a}}\cos\theta_{b\bar{b}}$ is falsified and the theory would be shown to be false. It is very possible that a different experimental setup where $\langle A(\bar{a})B(\bar{b})|A(a)B(b) \rangle$ takes a different form manages to violate the new equality mathematically and experimentally, but this would not be Bell violations of the form celebrated in the 2022 Physics Nobel Prize [5].

III. A LOCALLY REAL VIOLATION OF A BELL TEST

As in the quantum case, I will analyse the form that Bell gives for $\langle AB \rangle$ [6], see which classical axioms it rests on, and how to create a corrected Bell inequality. The purpose of this exercise is to certainly not abandon Locality, Realism, Determinability, and ideally not Measurement Integrity. That leaves suspicion for Instant Measurement and Measurement Independence. Although many would be content to take on the view that Instant Measurement is a matter of mathematical expediency, it is made redundant really by Measurement Integrity, in the sense that Instant Measurement only needs to be assumed for measurements that affect the outcome of future measurements, simply so that one can say that single measurements made are not so disruptive unto themselves as to become useless. Supposing Measurement Integrity, a measurement taken over some reasonable period of time would not be expected to disrupt itself as to require the additional supposition of Instant Measurement.

I find that there are two possible approaches. To abandon Measurement Independence and provide philosophical justification and a physical mechanism as to how it may be violated, or to find a new premise that is assumed in the Bell derivation that was previously missed. The former strategy is employed by Vervoort [7], and the latter by De Zela [8]. Although I would wish to explore both in this paper, I will explore only the findings of De Zela.

The primary strategy will be to demonstrate that $\langle AB \rangle = \int_{\Lambda} A(a, \lambda) \cdot B(b, \lambda) \cdot \rho(\lambda) d\lambda$ is too overconstrained a definition for the expected value of the product of the measurement outcomes as A and B (while they have measurement settings a and b). By the end of this section I will provide a more general form of the expectation, for which Bell violations that are known to occur will be no surprise or conundrum, and a new classical expectation that has yet to be known to be violated or violable.

Note first, that to test a Bell violation, we compare the experimentally observed simultaneous measurement outcomes at A and B, multiply those together in the order in which they are taken, and take a weighted average. That is if $A_{exp} = (1, 1, -1, 1, -1, \dots)$ and $B_{exp} =$

$(-1, -1, -1, 1, -1, \dots)$ then we have

$$\langle AB \rangle_{exp} = \sum_{A, B = \pm 1} A(a) \cdot B(b) \cdot P_{exp}(A(a) \cap B(b)) \quad (19)$$

With $P_{exp}(A(a) \cap B(b))$ equal to the number of times A(a) took on a certain value and B(b) took on a certain value divided by the number of measurements taken during the whole experiment.

If we define our theoretical probability measure, as is most commonly done, as:

$$P_{theo}(A \cap B) = \int_{\Lambda} \frac{1 + A(a, \lambda)}{2} \cdot \frac{1 + B(b, \lambda)}{2} \cdot \rho(\lambda) d\lambda \quad (20)$$

We do obtain:

$$\begin{aligned} \langle AB \rangle &= \sum_{A, B = \pm 1} A(a) \cdot B(b) \cdot P_{theo}(A(a) \cap B(b)) \quad (21) \\ &= \int_{\Lambda} A(a, \lambda) \cdot B(b, \lambda) \cdot \rho(\lambda) d\lambda \end{aligned}$$

Which is sufficient to arrive at the known Bell inequality that has been experimentally violated [16]. But in defining theoretical probability measure in this way we over constrain it. Because the measurement labels a and b appear only as labels, they have very little freedom to determine the outcomes at A and B, and in fact in this form, the locality assumption does make the theoretical probability measure take on the form of

$$P_{theo}(A \cap B) = \int_{\Lambda} p_{theo}(A|a, \lambda) \cdot p_{theo}(B|b, \lambda) \cdot \rho(\lambda) d\lambda \quad (22)$$

This separation is again sufficient to derive the violated Bell inequality. However, let us construct a probability measure that obeys the axioms of probability, obeys the predictions of quantum mechanics, is still classical in its mathematical nature, and uses the full fact that in this experiment the measurement settings a and b are vectors.

Consider a vector space V that contains an inner product, and a function f that is orthogonally additive and continuous. That is

$$\vec{u} \cdot \vec{v} = 0 \Rightarrow f(\vec{u} + \vec{v}) = f(\vec{u}) + f(\vec{v}) \quad (23)$$

By Gudder's Theorem [9], f must take the form:

$$f(\vec{v}) = c\vec{v} \cdot \vec{v} + \vec{k} \cdot \vec{v} \quad (24)$$

for some $c \in \mathbb{R}$ and $\vec{k} \in V$.

f is chosen to be orthogonally additive to satisfy the axioms of probability, and continuous so that small variations in the parameters of f lead to sensible variations in the value of f .

A probability measure is required to have the property $A \cap B = \emptyset \Rightarrow m(A \cup B) = m(A) + m(B)$

This will be achieved by having a probability measure that preferentially selects the vector $\vec{v}_\phi = (1, v_1, v_2, v_3) = (1, \hat{n}_\phi) \in V^4$. f_ϕ will have the following properties:

$$f_\phi(\vec{v}_\phi) = 1 \quad (25)$$

$$u = (1, -\hat{n}_\phi), w = (-1, \hat{n}_\phi) \Rightarrow f_\phi(\vec{u}_\phi) = f_\phi(\vec{w}_\phi) = 0 \quad (26)$$

$$\vec{v}_\psi = (1, \hat{n}_\psi) \Rightarrow f_\phi(\vec{v}_\psi) \in [0, 1] \quad (27)$$

These requirements on f_ϕ constrain it to the form of

$$f_\phi(\vec{v}_\psi) = \frac{1}{2}(1, \hat{n}_\phi) \cdot (1, \hat{n}_\psi) = \frac{1}{2}(1 + \hat{n}_\phi \cdot \hat{n}_\psi) \quad (28)$$

This is identical to Born's rule, the quantum probability that a system prepared in the state $|\phi\rangle$ will be measured to be in the state $|\psi\rangle$. That is:

$$f_\phi(\vec{v}_\psi) = \frac{1}{2}(1 + \hat{n}_\phi \cdot \hat{n}_\psi) = |\langle\psi|\phi\rangle|^2 \quad (29)$$

Not that the vectors $|\phi\rangle$ or $|\psi\rangle$ need to represent quantum states following this setup.

Consider now a pair of Particles prepared in the state represented by the vector $\phi_{AB} \in V^4 \otimes V^4$. If the measurement device A is prepared in the direction $\vec{a} \in V^3$, we write $\vec{\lambda}_A(\vec{a}) = \vec{a} \cdot \vec{\sigma}$ with $\vec{\sigma}$ being the vector of Pauli matrices. Identically for B . The result of this model is the correlation between measurement outcomes at measurement devices A and B is:

$$P_{theo}^{\phi_{AB}} = \langle \vec{\lambda}_A(\vec{a}) \otimes \vec{\lambda}_B(\vec{b}), \phi_{AB} \rangle \quad (30)$$

If, as in the analysis of the previous section, we prepare the entangled pair in the singlet state $\Psi_{AB}^- = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$, we arrive at the final result:

$$P_{theo}^{\phi_{AB}}(A(\vec{a}) \cap B(\vec{b})) = \frac{1}{4} \langle (1, A \cdot \vec{a}) \otimes (1, B \cdot \vec{b}), \phi_{AB} \rangle \quad (31)$$

$$P_{theo}^{\Psi_{AB}^-}(A(\vec{a}) \cap B(\vec{b})) = -\frac{1}{2} \vec{a} \cdot \vec{b} \quad (32)$$

The largest difference has been that [22](#) contains the product of two scalars to calculate an expected value, but [32](#) contains the inner product of two vectors. In this derivation, local realism was not broken, because the information required for Determinability is contained in the state of the particles that was imparted to both in their creation while they were in causal contact. The state of the two particles already contains in it the information of the correlation outcome even when the measurement settings are altered after separation.

This discussion and analyses is derivative of the work of De Zela [\[8\]](#). In the original paper, more time is gone into

exploring different ways of incorporating hidden variables into the above derivation that demonstrates the ease with which they fit, and also how superfluous they are in the derivation. All derivations using hidden variables return to

$$P_{theo}^{\Psi_{AB}^-}(A(\vec{a}) \cap B(\vec{b})) = -\frac{1}{2} \vec{a} \cdot \vec{b} \quad (33)$$

This formula is sufficient as well to recover the conditional expected values derived in the previous section in equation [17](#). The new Bell inequality, which in essence is equation [18](#) but with cosines replaced with dot products of normalised vectors, is mathematically inviolable. Thus concludes a derivation of quantum probabilities from deterministic, locally real premises.

IV. CONCLUSION AND THE CLASSICAL CASE

In the writing of this paper, it was hoped that I would arrive at a clear mechanism for how an experiment with classical objects could violate a Bell Inequality. The hope was that such a mechanism would remove all reason to conclude that a Bell violation entails non-local realism, as very few would suggest is the case of walking silicon droplets [\[10\]](#).

The reason for this lack of demonstration is that the mechanism found by the author was not sufficiently conclusively a description of Hydrodynamic Pilot Waves, but a potential description that takes on philosophical premises the author did not find persuading. I find it still an open question as to whether HQA will be able to violate a dynamic Bell test. The proposal for how this could be achieved is contained in Vervoort [\[7\]](#).

Instead, I opted to highlight that Bell's definition of expected value of the product of two measurement outcomes is highly constrained. Using a different probability measure than the one he provided, I provide a probability measure that is consistent with quantum mechanics and simultaneously determinism and local realism. Whether this in fact is the correct probability measure to be using is perhaps a more mathematical and philosophical discussion, but it does demonstrate that quantum probability calculations in no way require the abandonment of locality or realism.

ACKNOWLEDGMENTS

I give thanks to my professor and first mentor in theoretical physics, Professor John Bush. Further, I am grateful to every member of his lab, who despite my junior position and lack of familiarity in their fields of expertise, never treated me as anything less than an intellectual equal.

-
- [1] A. Aspect, *Nature* **398**, 189 (1999).
 - [2] A. Aspect, *Europhys* **2**, 73 (1991).
 - [3] D. Garisto, *Scientific American* (2022).
 - [4] B. Pena Cetto, .
 - [5] Z. Aspect, Clauser, .
 - [6] J. Bell, *Physics* **1**, 195 (1964).
 - [7] L. Vervoort, *European Physical Journal* **48** (2018).
 - [8] F. D. Zela, arxiv (2019).
 - [9] S. Gudder, *Stochastic Methods in Quantum Mechanics* (Dover Publications Inc., 1979).
 - [10] J. Bush, *Annual Review Fluid Mechanics* **47**, 269 (2015).